ARBITRAGE

Learning Outcome

- contrast the use of arbitrage and replication concepts in pricing forward commitments and contingent claims

An earlier lesson showed that riskless arbitrage opportunities arise if the "law of one price" does not hold—that is, an identical asset trades at different prices in different places at the same time. In the case of a derivative, we establish no-arbitrage conditions based on the payoff profile at maturity. As shown earlier, forward commitments with a *symmetric* payoff profile settle based on the difference between the forward price, $F_0(T)$, at contract inception ($t = 0$) and the underlying price, S_T, at contract maturity ($t = T$), or ($S_T - F_0(T)$) for a long forward position. For an underlying with no additional cash flows, it was shown that assets with a known future price must have a spot price equal to the future price discounted at the risk-free rate, r: $S_0 = S_T(1 + r)^{-T}$. For purposes of this lesson, we ignore any costs or benefits of owning the underlying other than the opportunity cost (or the risk-free rate, r).

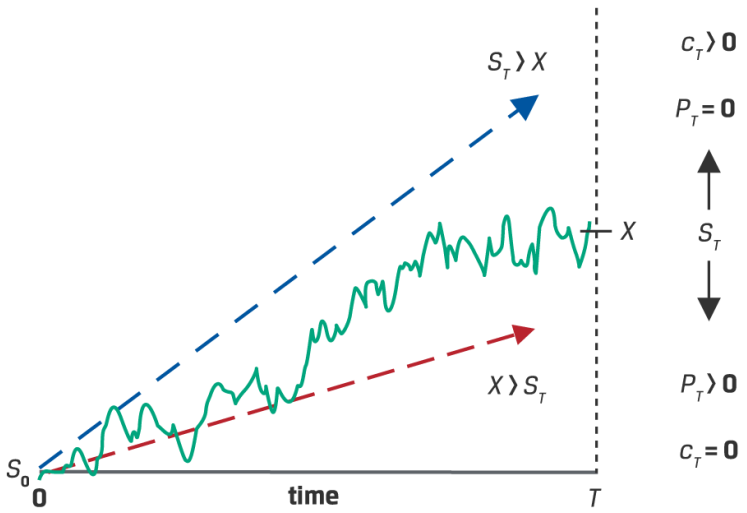
Contingent claims are characterized by *asymmetric* payoff profiles, introduced earlier. In the case of European options representing the right to purchase an underlying (or call option, c) or the right to sell an underlying (or put option, p) at a given exercise price (X) at maturity, the payoff profiles at maturity for an option buyer were shown to be:

$$c_T = \text{Max}(0, S_T - X) \quad (5)$$

$$p_T = \text{Max}(0, X - S_T) \quad (6)$$

Exhibit 3:

European Option Exercise at Expiration



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Recall that, unlike forward commitments with an initial price of zero, the option buyer pays the seller a premium (c_0 for a call and p_0 for a put at time $t = 0$), so the option buyer's *profit* at maturity was introduced earlier as follows (ignoring the time value of money):

$$\Pi = \text{Max}(0, S_T - X) - c_0 \quad (7)$$

Rate Your Confidence

High

Medium

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Pricing and Valuation of Options

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Equations 7 and 8 show the key distinction between forward commitments and contingent claims for purposes of arbitrage. The forward buyer enters into the contract with no cash paid upfront and has an unlimited gain or loss (bounded by zero for an underlying such as a stock that cannot have a negative price) at maturity as the underlying price rises or falls. The option buyer, in contrast, will exercise an option at maturity only if it is in the money. This *conditional* nature of option payoff profiles leads us to establish upper and lower no-arbitrage price bounds at any time *t*.

A call option buyer will exercise only if the spot price (S_T) exceeds the exercise price (X) at maturity. The *lower* bound of a call price is therefore the underlying's price minus the present value of its exercise price or zero, whichever is greater. In other words, an option which trades below its exercise value violates the no-arbitrage condition. The call buyer will not pay more for the right to purchase an underlying than the price of that underlying, which is the *upper* bound:

$$Max\left(0,\ S_t - X(1+r)^{-(T-t)}\right) < c_t \le S_t \qquad 9$$

$$c_{t,Lower\ bound} = Max\left(0,\ S_t - X(1+r)^{-(T-t)}\right) \qquad 10$$

$$c_{t,Upper\ bound=S_t} \qquad 11$$

A put option buyer will exercise only if the spot price, S_T, is below X at maturity. The exercise price, X, therefore represents the *upper* bound on the put value. The *lower* bound is the present value of the exercise price minus the spot price or zero, whichever is greater:

$$Max\left(0,\ X(1+r)^{-(T-t)} - S_t\right) < p_t \le X \qquad 12$$

$$p_{t,Lower\ bound} = Max\left(0,\ X(1+r)^{-(T-t)} - S_t\right) \qquad 13$$

$$p_{t,Upper\ bound} = X \qquad 14$$

EXAMPLE 4

Call Option Upper and Lower Bounds

Consider a one-year call option with an exercise price, X, of EUR 1,000. The underlying asset, S_0, trades at EUR 990 at time $t = 0$ and the risk-free rate, r, is 1%. What are the no-arbitrage upper and lower bounds in six months' time if the underlying asset price, S_t, equals EUR 1,050?

As the option buyer will exercise only if $S_T > X$ at $t = T$, the *lower* bound is equal to S_t − PV (X) or zero, whichever is greater:

$$c_{t,Lower\ bound} = Max\left(0,\ S_t - X(1+r)^{-(T-t)}\right)$$

$$c_{t,Lower\ bound} = Max(0,\ EUR\ 1{,}050 - EUR\ 1{,}000(1.01)^{-0.5})$$

$$c_{t,Lower\ bound} = Max(0,\ EUR\ 54.96)$$

The call buyer will not pay more than S_t for the right to purchase the underlying:

$$c_{t,Upper\ bound} = S_t$$

$$c_{t,\ Upper\ bound} = EUR\ 1{,}050$$

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GA
Shoudn't the upper bound for a call option be the exercise value? as it is the "limit" the investor will pay?

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LB

Should not the upper limit of the put option price be the discounted strike price? That is, p_t <= X/(1+r)^(1/(T-t))

Created 5 months ago by Lorenzo Berti

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